

# Nicolas Dufour *Senior Staff Platform Engineer*

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A creative and motivated engineer with more than 20 years of experience in development, integration, and system/network administration, both on-premise and cloud platforms.

## Skills

|                                                                        |           |                                                                      |           |
|------------------------------------------------------------------------|-----------|----------------------------------------------------------------------|-----------|
| <b>Infrastructure as Code</b><br>terraform   ansible                   | ● ● ● ● ● | <b>Container orchestration</b><br>kubernetes   helm   docker         | ● ● ● ● ● |
| <b>AI</b><br>claude-code   droid   agents   hermes agent               | ● ● ● ● ● | <b>CI/CD   GitOps</b><br>git   github actions   gitlab   flux        | ● ● ● ● ● |
| <b>Scripting</b><br>Bash   Powershell                                  | ● ● ● ● ● | <b>Monitoring</b><br>grafana   prometheus   ELK   splunk             | ● ● ● ● ● |
| <b>Programming languages</b><br>Go   Python   Rust   Java   C   Erlang | ● ● ● ● ● | <b>Cloud computing platforms</b><br>AWS   Azure   GCP   DigitalOcean | ● ● ● ● ● |
| <b>Databases</b><br>MySQL   Postgresql   Oracle   Aura                 | ● ● ● ● ● |                                                                      |           |

## Professional Experience

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                      |
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| <b>Senior Staff Platform Engineer, Nav Technologies</b><br>Bringing my experience in GitOps and SRE to improve both incident and operation flows: <ul style="list-style-type: none"><li>Improving in-house tooling, along with AI tooling</li><li>Deployed custom grafana/prometheus along side splunk and helped dev teams to better monitor their applications</li><li>Leading SRE practices within the company</li><li>Setting up a process to create kubernetes clusters faster and automated for both ephemeral environments and production</li><li>Teaching/mentoring the engineering team on good practices with kubernetes</li><li>Leading the migration process for all kubernetes clusters of the company</li><li>Lead the migration from Gitlab to Github</li><li>Setting up a rock solid ArgoCD based pipeline for kubernetes and educating the engineering team to use it next to Github</li><li>Lead the DR process with several levels</li></ul> | 08/2023 – present<br>West Orange, NJ |
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**Senior SRE, Sectigo Inc.**

10/2021 – 06/2023

Roseland, NJ

Worked among a tight-knit team of SREs brought together to improve the organization's services and its related infrastructure including:

- designed and deployed several Certificate Transparency services to help foster a safer community for certificate issuance among CA authorities
- designed and deployed workflows to allow automatic issuance of certificates (90 days initiative)
- managed and set up on-premise VMs (Proxmox, VMware)
- created and managed Kubernetes clusters and their workloads using
  - kustomizations
  - flux v2 / gitops principles
  - helm templates
- managed the infrastructure configuration via Ansible/terraform and Puppet, and created new Ansible roles

**Senior DevOps Architect, Rivet Logic Corporation**

01/2016 – 10/2021

West Orange, NJ

- Designed multiple SAML authentication systems with different IDPs and SPs (primarily Apache-based).
- Helped Marriott migrate several projects to the cloud and maintained those services over the years (mostly CMS based systems)
- Created automation involving CloudFormation, Terraform, Ansible, and AWS resources to ensure the process is repeatable and reliable.
- Co-lead the MS (Managed Services) team.
- Extensive AWS experience across most services (VPC, EC2, S3, RDS, Lambda, CloudWatch, SNS, etc).
- Introduced and taught Kubernetes in the company.
- Designed and deployed Kubernetes cluster on AWS using both Kops and EKS.
- Performed multiple Lift-and-shift in AWS.
- Helped design monitoring solutions using both in-house and outside of AWS.

**Software Architect, Rivet Logic Corporation**

10/2010 – 01/2016

West Orange, NJ

- Started the development of a brand new marketing site for MasterCard by customizing Alfresco.
- Helped to design and customize Alfresco for the NFL public site.
- Developed and customized Liferay site for IEEE and Noridian.
- Led the development and maintenance of 3 marketing sites for MasterCard.

**R&D Engineer, IPsoft**

11/2006 – 10/2010

New York, NY

- Key developer for IPsoft's proprietary ITSM software: IPcenter.
- Designed and developed a new AS400 system monitoring tool (Java/JTOpen/Groovy/Netty).
- Designed and developed an event correlation engine (Java/Esper/ActiveMQ/Hibernate).
- Implemented several "eBonding" system integrations (Java/ActiveMQ/Hibernate).
- Extends IPsoft's current framework to connect with new customers.

**Senior Software Architect, Time Inc**

01/2007 – 09/2009

New York, NY

- Designed and developed a new Content Management System (CMS) managing magazine. websites for Real Simple, Coastal Living, Southern Living, Cooking Light, Southern Accents, and others. TIPS v1.0 (Java/Spring/JCR/Flex/Ajax).
- Led the CMS development team to TIPS v3.0 (Java/Spring/JCR/Flex/LCDS).

**Linux Team Leader, NeoMetSys**

05/2004 – 09/2009

Paris, France

- Work for the European Commission.
  - Integration of an American Air Traffic Simulator for the European environment.
- Work in partnership with Metron Aviation Inc for NASA and FAA.
  - Production of a new Simulation/Analysis tool for NASA.
  - Software wrapper developed between FACET and POET.
  - Extensive use of different database sources (Oracle/MS SQL/ Derby).
- Work in partnership with South Africa Aviation Administration.
  - Designed modules of the new Flight Data Repository for the South Africa Aviation Administration.
- Internal projects.
  - Adapted the primary American airline/airport collaboration tool for the European environment.
  - Led the Linux team and provided support on internal projects.

**Consultant, NeoSys**

02/1999 – 05/2004

Paris, France

- Set up and maintained the company network and appliances according to users' wishes.
- Wrote a comparative study of present Web middleware in the USA in 2003.
- Managed the primary simulation tool of the Air Traffic Flow research team (TACT) at Eurocontrol for 3 years.

 **Languages**

|         |           |        |           |
|---------|-----------|--------|-----------|
| English | ● ● ● ● ● | French | ● ● ● ● ● |
| German  | ● ● ● ● ● |        |           |

 **Education****Computer/Geoscience Master, Université de Paris 6**1995 – 1998  
Paris, France**Geochemistry/Geophysics Bachelor, Université de Paris 7**1993 – 1995  
Paris, France**Biology Associate, Université de Paris 7**1991 – 1993  
Paris, France

## Certificates

**Amazon Web Services Cloud Practitioner**

2020 - 2023

**AWS Technical Professional Online**

2017 - 2023

## Projects

**Home Datasensing**, *Home datacenter cluster and wireless sensors*

2010 – present

On-going project to create a compact datacenter and wireless sensor network for home usage.

**Refuge**, *Distributed storage solution*

2011 – 2013

Project dedicated at exploring distributed solutions to increase privacy for data storage/sharing.